

Yusuf Yiğit PİLAVCI

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🌐 Website
🎓 Scholar
🐙 Gitlab
🐙 Github



Education and Employment

- 2026 – present 📌 **Maître de Conférences**, Université de Caen Normandie
Research team: IMAGE/GREYC
Teaching establishment: IUT/BUT Mesures Physiques
- 2025 – 2026 📌 **Postdoctoral Researcher**, IMT Nord Europe
Generalization of the detectors for AI-generated images.
Supervised by Vincent Itier.
- 2023 – 2025 📌 **Postdoctoral Researcher**, CRISTAL Lab
Design of geometrical priors for inverse problems of bivariate signals.
Supervised by Pierre-Antoine Thouvenin, Jérémie Boulanger and Pierre Chainais.
- 2019 – 2022 📌 **PhD Degree**, Université Grenoble Alpes, GIPSA Lab.
Thesis title: *Wilson's Algorithm for Randomized Linear Algebra*
Supervised by Nicolas Tremblay, Simon Barthelmé and Pierre-Olivier Amblard.
Defended on Nov. 2022.
- 2017 – 2019 📌 **M.Sc. in Computer Science and Engineering**, Politecnico di Milano
Final Grade: 108.0 / 110
Thesis: *Random Spanning Forests, Theory and Applications*
- 2012 – 2017 📌 **Bachelor's degree in Electrical and Electronics Engineering** in Middle East Technical University.
CGPA: 3.79 / 4.00 (10th/375)
- 2014 – 2017 📌 **Minor degree in Computer Engineering** in Middle East Technical University.

Research Publications

Journal Articles

- 1 Y. Y. Pilavcı, E. T. Güneyi, C. Cengiz, and E. Vural, "Graph domain adaptation with localized graph signal representations," *Pattern Recognition*, p. 110 628, 2024.
- 2 Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, "Graph tikhonov regularization and interpolation via random spanning forests," *IEEE Transactions on Signal and Information Processing over Networks*, 2021.
- 3 M. Turan, Y. Y. Pilavcı, I. Ganiyusufoglu, H. Araujo, E. Konukoglu, and M. Sitti, "Sparse-then-dense alignment-based 3d map reconstruction method for endoscopic capsule robots," *Machine Vision and Applications*, vol. 29, no. 2, pp. 345–359, 2018.

International Conference Proceedings

- 1 N. Cigercioglu, Y. Pilavci, and H. Guney-Deniz, "Pos1490-hpr can knee osteoarthritis be predicted using machine learning with pedobarographic data," in *Annals of the Rheumatic Diseases*, vol. 84, Elsevier, 2025, pp. 1487–1488.
- 2 Y. Y. PİLAVCI, P. Palud, J. Flamant, P.-A. Thouvenin, J. Boulanger, and P. Chainais, "Time and covariance smoothing for restoration of bivariate signals," in *2025 IEEE Statistical Signal Processing Workshop (SSP)*, IEEE, 2025, pp. 141–145.

- 3 Y. Y. Pilavcı, J. Boulanger, P.-A. Thouvenin, and P. Chainais, "Denoising bivariate signals via smoothing and polarization priors," in *2024 32nd European Signal Processing Conference (EUSIPCO)*, IEEE, 2024, pp. 2602–2606.
- 4 Y. Y. Pilavcı, P.-O. Amblard, S. Barthelmé, and N. Tremblay, "Variance reduction in stochastic methods for large-scale regularised least-squares problems," in *30th European Signal Processing Conference, (EUSIPCO)*, 2022.
- 5 Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, "Smoothing graph signals via random spanning forests," in *ICASSP 2020-2020 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2020, pp. 5630–5634.
- 6 Y. Y. Pilavcı and N. Farrugia, "Spectral graph wavelet transform as feature extractor for machine learning in neuroimaging," in *ICASSP 2019-2019 IEEE International Conference on Acoustics, Speech and Signal Processing (ICASSP)*, IEEE, 2019, pp. 1140–1144.

National Conference Proceedings

- 1 Y. Y. Pilavcı, J. Flamant, P.-A. Thouvenin, J. Boulanger, and P. Chainais, "An admm algorithm to restore bivariate signals with joint time and covariance regularization," in *GRETSI 2025 - Colloque Francophone de Traitement du Signal et des Images*, 2025.
- 2 Y. Y. Pilavcı, P.-O. Amblard, S. Barthelme, and N. Tremblay, "Variance reduction for inverse trace estimation via random spanning forests," in *GRETSI 2022 - Colloque Francophone de Traitement du Signal et des Images*, Nancy, France, 2022.

Teaching Experience

2023 – 2024	 IMT Nord Europe Security, Authentication and Forensics (SAFE), Compression and error-correcting codes (CODES), Signal theory
2024 – 2025	 IMT Nord Europe Signal theory, Tools for data science  Centrale Lille Advanced algorithms and programming  IMT Nord Europe Signal theory
2025 – 2026	 IMT Nord Europe Signal theory
2026 –	 IUT Grand Ouest Normandie, BUT Mesure Physique Informatics

Skills

Languages	 Turkish (Maternal), English (Fluent), French (Intermediate)
Software	 Python, Julia, C/C++, Matlab, $\text{\LaTeX}$
Qualifications	 Qualified in sections 26, 27 and 61 of CNU